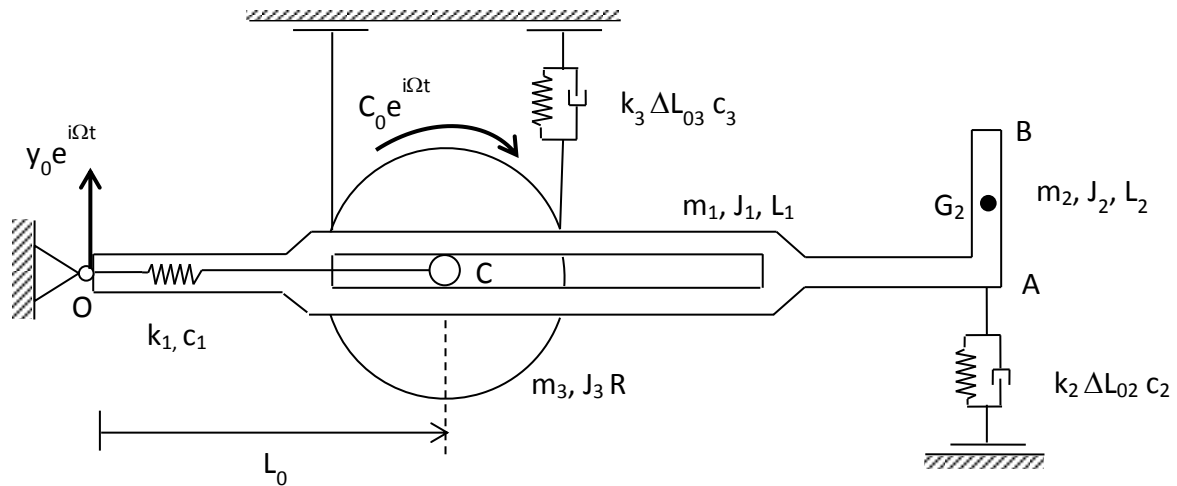


Mechanical Systems Dynamic

December 12th, 2015 – Version B



$m_1 = 10 \text{ kg}$	$k_1 = 1000 \text{ N/m}$	$y_{02} = 0.01 \text{ m}$
$m_2 = 2 \text{ kg}$	$k_2 = 2000 \text{ N/m}$	$y_{03} = 0.02 \text{ m}$
$m_3 = 5 \text{ kg}$	$k_3 = 3000 \text{ N/m}$	$C_{01} = 100 \text{ Nm}$
$J_1 = 0.8 \text{ kgm}^2$	$\Delta L_{02} = 0.1 \text{ m}$	$\varphi_1 = \pi/6 \text{ rad}$
$J_2 = 0.01 \text{ kgm}^2$	$\Delta L_{03} = 0.0828 \text{ m}$	$\varphi_2 = -\pi/6 \text{ rad}$
$J_3 = 0.2 \text{ kgm}^2$	$c_1 = 2 \text{ Ns/m}$	$\varphi_3 = \pi/3 \text{ rad}$
$L_1 = 1.0 \text{ m}$	$c_2 = 4 \text{ Ns/m}$	$\Omega_1 = 28 \text{ rad/s}$
$L_2 = 0.3 \text{ m}$	$c_3 = 6 \text{ Ns/m}$	$\Omega_2 = 14 \text{ rad/s}$
$L_0 = 0.6 \text{ m}$		$\Omega_3 = 7 \text{ rad/s}$
$R = 0.3 \text{ m}$		

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by a “L shaped” rigid body (the center of gravity of the horizontal part is in the middle of the horizontal bar of length L_1 , namely at $L_1/2$ from point O) and by a disk which is sliding and rotating with respect to the first body. The disk center is connected with point O with a spring and damper system (k_1 c_1). The static preload of the springs k_2 and k_3 is given and is equal to ΔL_{02} and ΔL_{03} respectively. In the static configuration the distance between the center of the disk and point O is L_0 .

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the torque $C_0 e^{i\Omega t}$ at the disk. Output: the absolute rotation of the disk and the vertical displacement of point C. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the vertical displacement $y_0 e^{i\Omega t}$ to the hinge O. Output: the horizontal displacement of point C and the force transmitted by the spring and damper system (k_3 c_3). Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the time history of the vertical force applied to the hinge O, given the following inputs: the vertical periodic displacement $y(t)$ defined by the equation below and a harmonic torque $C_{01} e^{i(\Omega_1 t + \varphi_1)}$ applied to the disk. The time history shall be defined for one period of oscillation with a time step of 0.001 s. Save the results as SNxxx5.

$$y(t) = y_{02} \cos(\Omega_2 t + \varphi_2) + y_{03} \cos(\Omega_3 t + \varphi_3)$$

6. Compute the static preload of the k_3 spring applying a static preload of the k_2 spring two times of the given ΔL_{02} and keeping the same static configuration of the system. Write the numerical result on the sheets of the solution.

Note: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select one of the following of the folders *Test Dec. 12 2015 L07- B, Test Dec. 12 2015 L14- B, Test Dec. 12 2015 L15- B* according to the classroom and test version
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse...* (or *Sfoglia...*)
7. Fill the *Title* field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.